ANALYTICAL CUMULATIVE EXAM

February 10, 1996

A common method to determine the concentration of an unknown is standard addition. A small amount of standard is added to the solution containing the unknown and the analysis is then carried out. How the data is treated depends on how the samples were prepared before analysis. In this cume you will be asked to develop data treatments for two different procedures for standard addition.

Method I

Consider the determination of Li by atomic emission in which the standard contained 1.62 ppm Li. 10 mL aliquots of unknown solution were placed in each of five 100 mL volumetric flasks, different volumes of standard were then added and the flasks diluted to 100 mL. The following table resulted.

Unknown V_x (mL)	Standard V_s (mL)	Final Volume V_T(mL)	Emission intensity I (Arbitrary units)
10.00	0.00	100.0	309
10.00	5.00	100.0	452
10.00	10.00	100.0	600
10.00	15.00	100.0	765
10.00	20.00	100.0	906

1. Derive the theoretical equation that shows a straight line relation between emission intensity (I) to the volume of standard added (V_s). You may assume that emission intensity is directly proportional to lithium concentration. Let Vx be the concentration of the unknown, V_T the total volume.

2. Show that when the line is extrapolated back to zero emission the concentration of the unknown in the 10 mL aliquot is equal to:

$$C_x = -C_s \frac{V_s}{V_x}$$

   Where Vs is taken from the graph at I = 0

3. Determine the unknown lithium concentration in the 10 mL aliquot by the graphical procedure. Use graph paper provided.

Method II

Consider another procedure in which an unknown sample of Cu gave an absorbance of 0.262 (A_x) in an atomic absorption analysis. Then 1.00 mL (V_s) of a standard solution containing 100 ppm copper (C_s) was mixed with 95 mL of the unknown (V_x), then further diluted to 100 mL (V_T) the absorbance read 0.500 (A_{x+s})

1. Assuming that absorbance is directly proportional to the copper concentration, derive the formula:

$$\frac{A_x}{A_{s+x}} = \frac{C_x}{\frac{C_x V_x}{V_T} + \frac{C_s V_s}{V_T}}$$

2. Determine the concentration of Cu in the unknown in ppm.